the duel, the maths behind what does work

numbers are the films own unless i say its a guess

1. the reactor charging the flywheels

the orbital ship has a 14 MW reactor that fills the flywheels over a day

$$E = Pt = 14 \times 10^6 \times 24 \times 3600 = 14 \times 10^6 \times 86{,}400 = 1.21 \times 10^{12}\,\text{J} = 1.2\,\text{TJ}$$

matches the films 1.2 TJ exactly. the reactor makes 140 MW of heat for 14 MW of power so 126 MW has to go out the radiators. ε is emissivity, σ is the Stefan Boltzmann constant, fins radiate off both sides

$$\begin{aligned} & q = 2\varepsilon\sigma T^4 \\ \text{at } 800\,\text{K}: \quad & q = 2 \times 0.9 \times 5.67 \times 10^{-8} \times 800^4 = 41{,}800\,\text{W/m}^2 \\ & A = \frac{126 \times 10^6}{41{,}800} = 3010\,\text{m}^2 \\ \text{at } 600\,\text{K}: \quad & A = \frac{126 \times 10^6}{13{,}200} = 9530\,\text{m}^2 \end{aligned}$$

so 55 by 55 m up to about 100 by 100 m of fins, easy on a 1.7 km ship. only thing a bit high is 10% conversion, real thermoelectrics do 5 to 8%

2. the flywheels

56 wheels at 3000 rpm. radius is my guess, 3 m

$$\begin{aligned} E_{\text{wheel}} &= 1.2 \times 10^{12}/56 = 21.4\,\text{GJ} \\ \omega &= 3000 \times 2\pi/60 = 314\,\text{rad/s}, \qquad v_{\text{rim}} = \omega r = 314 \times 3 = 942\,\text{m/s} \\ m &= \frac{2E}{v^2} = \frac{2 \times 21.4 \times 10^9}{942^2} = 48{,}300\,\text{kg} \end{aligned}$$

hoop stress in a spinning rim, carbon fibre at 1600 kg/m^3

$$\sigma = \rho v^2 = 1600 \times 942^2 = 1.42\,\text{GPa}$$

carbon fibre composite holds 2 to 3 GPa round the hoop so thats a real safety margin. half spin one way half the other so the spin cancels, I is moment of inertia, L angular momentum

$$\begin{aligned} I &= mr^2 = 48{,}300 \times 9 = 434{,}700\,\text{kg m}^2 \\ L_{\text{one}} &= I\omega = 434{,}700 \times 314 = 1.37 \times 10^8 \\ L_{\text{total}} &= 28L - 28L = 0 \end{aligned}$$

if they all spun the same way thats $56 \times 1.37 \times 10^8 = 7.7 \times 10^9$ and turning the ship at 0.01 rad/s would take $\tau = L\Omega = 7.7 \times 10^7\,\text{N m}$ fighting it the whole time. the film got that right

3. the orbital gun shot

rod is tungsten, 0.3 m across and 18 m long, barrel about 1016 m

$$\begin{aligned} m &= \rho\pi r^2 L = 19300 \times \pi \times 0.15^2 \times 18 = 24{,}560\,\text{kg} \\ KE &= \tfrac{1}{2}mv^2 = 0.5 \times 24{,}560 \times 5000^2 = 307\,\text{GJ} = 73\,\text{t TNT} \\ \eta &= 307/600 = 51\% \end{aligned}$$

all match the film. the long barrel keeps it gentle

$$\begin{aligned} a &= \frac{v^2}{2L} = \frac{5000^2}{2 \times 1016} = 12{,}300\,\text{m/s}^2 = 1250\,g \\ t &= \frac{2L}{v} = \frac{2032}{5000} = 0.41\,\text{s} \\ P &= \frac{0.6 \times 10^{12}}{0.41} = 1.5\,\text{TW} \end{aligned}$$

the sabot grips the rod down its whole length so the load goes in as shear

$$F = ma = 24{,}560 \times 12{,}300 = 3.0 \times 10^8\,\text{N}, \qquad \tau = \frac{F}{\pi dL} = \frac{3.0 \times 10^8}{\pi \times 0.3 \times 18} = 17.8\,\text{MPa}$$

tungsten wont even notice that, compare 26 GPa on the destroyers round

4. the destroyers round going up

to coast up to 380 km in a vacuum it needs

$$\tfrac{1}{2}v^2 = \mu\left(\tfrac{1}{R} - \tfrac{1}{r}\right) = 3.986 \times 10^{14} \times 8.835 \times 10^{-9} = 3.52 \times 10^6 \quad \Rightarrow \quad v = 2650\,\text{m/s}$$

it has 7800 so its got plenty. leaving the air at 6.9 km/s at 80 km it gets to 380 km at

$$v^2 = 6900^2 - 2\mu\left(\tfrac{1}{6451\,\text{km}} - \tfrac{1}{6751\,\text{km}}\right) = 4.76 \times 10^7 - 5.49 \times 10^6 \quad \Rightarrow \quad v = 6490\,\text{m/s}$$

nose heating at sea level (Sutton Graves), r_n is nose radius, guess 3 mm

$$q = k\sqrt{\rho/r_n}\,v^3 = 1.74 \times 10^{-4} \times \sqrt{1.225/0.003} \times 7800^3 = 1.7 \times 10^9\,\text{W/m}^2$$

tungsten has to be heated to 3695 K and melted before it comes off

$$H = c\Delta T + L_f = 134 \times 3400 + 192{,}000 = 0.65\,\text{MJ/kg}$$

air density falls off with height ($H_s = 8.5$ km) and it climbs at $7800 \sin 70° = 7330$ m/s, so the air it goes through adds up to

$$\int \sqrt{\rho}\, dt = \sqrt{\rho_0}\,\frac{2H_s}{v_z} = 1.107 \times \frac{17{,}000}{7330} = 2.57$$

$$\text{nose lost} = \frac{kv^3}{\sqrt{r_n}} \times \frac{2.57}{\rho_w H} = \frac{1.51 \times 10^9 \times 2.57}{19300 \times 0.65 \times 10^6} = 0.31\,\text{m}$$

thats the worst case, the nose blunts as it melts which cuts the heating, so 10 to 30 cm off a 2 m rod. it survives and the white hot sheath in the film is right. nothing burns once its out of the air

5. the rod coming down and going in

β is ballistic coefficient, C_d about 0.15

$$\beta = \frac{m}{C_d A} = \frac{24{,}560}{0.15 \times \pi \times 0.15^2} = 2.3 \times 10^6\,\text{kg/m}^2$$

thats so heavy for its size the air barely touches it. same heating sum as above with a blunter 5 cm nose at 5 km/s

$$\int \sqrt{\rho}\, dt = 1.107 \times \frac{17{,}000}{5000 \sin 44^\circ} = 1.107 \times \frac{17{,}000}{3473} = 5.42$$

$$\text{nose lost} = \frac{1.74 \times 10^{-4} \times 5000^3}{\sqrt{0.05}} \times \frac{5.42}{19300 \times 0.65 \times 10^6} = 0.04\,\text{m}$$

4 cm off 18 m so it hits whole. then in the water a long rod wears away as it goes, L is rod length

$$P = L\sqrt{\rho_{\text{rod}}/\rho_{\text{water}}} = 18 \times \sqrt{19.3/1.025} = 18 \times 4.34 = 78\,\text{m}$$

so it really does dive about 80 m and eat itself to nothing like the film shows, and dumping 73 t of TNT worth of energy right under the keel breaks the destroyer